# Portland State University

DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING

## ECE332

## LAB 3: WAVEGUIDES

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# 1 Pre-lab assignment

**1.** Which of the following types of transmission lines does NOT support the propagation of transverse-electromagnetic (TEM) waves? (include all that apply.)

(a) Rectangular Waveguides

(b) Coaxial Cable

(c) Parallel-plate Lines

(d) Optical Fiber

**2.** What is the dominant mode for transverse magnetic (TM) waves propagating through a waveguide?

(a) $TM_{00}$

(b) $TM_{10}$

(c) $TM_{01}$

(d) $TM_{11}$

(e) None of the above

**3.** What is the dominant mode for the transverse electric (TE) waves propagating through a waveguide?

(a) $TE_{00}$

(b) $TE_{01}$

(c) $TE_{10}$

(d) $TE_{11}$

(e) None of the above

**4.** What type of filter does a resonant cavity implement?

(a) Low Pass

(b) High Pass

(c) Notch

(d) Bandpass

(e) None of the above

**5.** Which instrument is used for measuring the Standing Wave Ratio (SWR) if the signal is propagating through a waveguide?

(a) Systron/Donner 5000

(b) Hewlett Packard 415E

(c) Neither, you made those up!

# 2 Introduction

This lab investigates how electromagnetic waves travel inside rectangular waveguides, with a focus on propagation behavior, cutoff conditions, and the formation of standing wave patterns. Waveguides are hollow metal structures used to guide microwave-frequency signals, and unlike coaxial lines they cannot support TEM modes. Instead, they operate using transverse electric (TE) and transverse magnetic (TM) modes, each with a specific cutoff frequency determined by the waveguide's physical dimensions.

The word "transverse" means the field has no component along the direction of propagation, which we call the $z$-axis. For TE modes the electric field is all transverse and only the magnetic field has a $z$-component, and for TM modes it is the other way around. The subscripts $m$ and $n$ count how many half-wavelength variations of the field fit across the guide width $a$ (along $x$) and height $b$ (along $y$). For example $TE_{10}$ has one half-wavelength of E-field variation across the width and no variation across the height. The cutoff frequency for any $(m, n)$ mode is written $f_{mn}$, and it sets the minimum frequency where that mode can actually propagate.

In this experiment, a slotted waveguide was used to visualize standing waves created by reflections from a short-circuit load. By sliding a probe along the slot, we measured changes in the electric field and identified the positions of maxima and minima. These measurements allowed us to determine the guide wavelength $\lambda_g$, as well as calculate phase and group velocities at different operating frequencies. Overall, the lab emphasizes how boundary conditions and waveguide geometry shape wave propagation and how these effects differ from ordinary free-space transmission.

# 3 Part 1: Wavelength Measurements

### Objective

The goal of this experiment is to study how electromagnetic waves travel inside a rectangular waveguide and to measure the guide wavelength ($\lambda_g$) at several microwave frequencies. Using a slotted waveguide and an SWR meter, we observe the standing wave pattern created by a short-circuit termination. From the measured minima spacing, we calculate the guide wavelength, phase velocity ($v_p$), and group velocity ($v_g$) for the dominant $TE_{10}$ mode.

### Theory

A rectangular waveguide is a hollow metal structure that carries electromagnetic waves only when the operating frequency is above a certain cutoff value. For the dominant $TE_{10}$ mode, the cutoff wavelength is $\lambda_c = 2a/m$ where $a$ is the wider dimension of the waveguide and $m = 1$ for the $TE_{10}$ mode.

The relationship between the guide wavelength $\lambda_g$, the free-space wavelength $\lambda$, and the cutoff wavelength $\lambda_c$ is:

$$\lambda_g = \frac{\lambda}{\sqrt{1 - (\lambda/\lambda_c)^2}}$$

Inside the waveguide, the wave travels with a phase velocity and group velocity given by:

$$v_p = \frac{c}{\sqrt{1-(f_c/f)^2}}, \qquad v_g = c\sqrt{1-(f_c/f)^2}$$

Here $f_c$ is the cutoff frequency and $f$ is the operating frequency. Their product remains constant: $c^2 = v_p v_g$. This means that as the phase velocity becomes greater than the speed of light, the group velocity (which carries energy) decreases at the same rate, keeping the physics consistent.

## Setup

The experiment uses a slotted rectangular waveguide connected to a Systron-Donner 5000 oscillator operating from 8–12.4 GHz. A movable probe inside the slotted line detects the electric field intensity, and an HP 415E SWR meter displays the measured signal.

**Steps:**

1. Attach a short-circuit plate to the end of the waveguide.
2. Connect the probe output to the SWR meter using a BNC cable.
3. Set the oscillator to the desired test frequency (e.g., 11 GHz).
4. Adjust the probe depth to get a strong reading without disturbing the field.
5. Slide the probe along the slot to find the maxima and minima of the standing wave.
6. Record the positions of at least three minima or maxima to calculate $\lambda_g/2$.

## Equations Used

Waveguide Wavelength: $\lambda_g = 4\Delta d = 4(d_{n+1} - d_n)$
Phase Velocity: $v_p = f\lambda_g$
Group Velocity: $v_g = c^2/v_p$, where $c = 3 \times 10^8$ m/s (speed of light in vacuum)

## Results

| $f = 11$ **GHz** | | | | |
|---|---|---|---|---|
| | Position (mm) | $\lambda_g$ (mm) | $v_p$ (m/s) | $v_g$ (m/s) |
| Min 1 | 170.1 | 31.2 | $3.432 \times 10^8$ | $2.622 \times 10^8$ |
| Max 1 | 162.3 | 36.0 | $3.960 \times 10^8$ | $2.273 \times 10^8$ |
| Min 2 | 153.3 | 32.4 | $3.564 \times 10^8$ | $2.525 \times 10^8$ |
| Max 2 | 145.2 | 34.4 | $3.784 \times 10^8$ | $2.378 \times 10^8$ |
| Min 3 | 136.6 | 29.6 | $3.256 \times 10^8$ | $2.764 \times 10^8$ |
| Max 3 | 129.2 | N/A | N/A | N/A |
| Avg: | | 32.7 | $3.599 \times 10^8$ | $2.501 \times 10^8$ |

| $f = 9.5$ **GHz** | | | | |
|---|---|---|---|---|
| | Position (mm) | $\lambda_g$ (mm) | $v_p$ (m/s) | $v_g$ (m/s) |
| Min 1 | 174.4 | 44.0 | $4.180 \times 10^8$ | $2.153 \times 10^8$ |
| Max 1 | 163.4 | 43.6 | $4.142 \times 10^8$ | $2.173 \times 10^8$ |
| Min 2 | 152.5 | 42.0 | $3.990 \times 10^8$ | $2.256 \times 10^8$ |
| Max 2 | 142.0 | 42.4 | $4.028 \times 10^8$ | $2.234 \times 10^8$ |
| Min 3 | 131.4 | 43.6 | $4.142 \times 10^8$ | $2.173 \times 10^8$ |
| Max 3 | 120.5 | N/A | N/A | N/A |
| Avg: | | 43.1 | $4.096 \times 10^8$ | $2.197 \times 10^8$ |

| $f = 8.5$ **GHz** | | | | |
|---|---|---|---|---|
| | Position (mm) | $\lambda_g$ (mm) | $v_p$ (m/s) | $v_g$ (m/s) |
| Min 1 | 166.4 | 62.4 | $5.304 \times 10^8$ | $1.697 \times 10^8$ |
| Max 1 | 150.8 | 47.6 | $4.046 \times 10^8$ | $2.224 \times 10^8$ |
| Min 2 | 138.9 | 64.4 | $5.474 \times 10^8$ | $1.644 \times 10^8$ |
| Max 2 | 122.8 | 46.0 | $3.910 \times 10^8$ | $2.302 \times 10^8$ |
| Min 3 | 111.3 | 58.0 | $4.930 \times 10^8$ | $1.826 \times 10^8$ |
| Max 3 | 96.8 | N/A | N/A | N/A |
| Avg: | | 55.7 | $4.733 \times 10^8$ | $1.902 \times 10^8$ |

The tables above show the waveguide SWR minimums and maximums.

Measured and recorded dimensions of $a$ and $b$ of the waveguide:
$a$ (width) = 22.92 mm, $b$ (height) = 10.29 mm.

Distance from the short circuit termination to the nearest detectable minimum (used in Q3/Q4):

- 11 GHz: 83.5 mm
- 9.5 GHz: 95.6 mm
- 8.5 GHz: 85.4 mm

## Discussion

From the measurements, the guide wavelength ($\lambda_g$) becomes shorter as the operating frequency increases. For example, $\lambda_g$ decreased from about 55.7 mm at 8.5 GHz to 32.7 mm at 11 GHz. This agrees with waveguide theory, which predicts that higher frequencies produce shorter wavelengths inside the guide. The calculated phase velocities $v_p$ are greater than the speed of light, which is normal for waveguides because phase velocity does not represent the speed of energy flow. Meanwhile, the group velocity $v_g$ (the actual energy transfer speed) remains below $c$ and gets smaller as frequency decreases. This opposite behavior between $v_p$ and $v_g$ satisfies the condition $v_p v_g = c^2$. Overall, the results match the expected behavior of a rectangular waveguide operating in the $TE_{10}$ dominant mode. Minor differences in the measured maxima and minima could be due to probe positioning errors or small inaccuracies in the SWR meter, but the overall trends remain consistent with theory.

## 4 Part 2: Frequency Measurements

### Objective

The goal of this part of the experiment is to measure the resonance frequencies of electromagnetic waves in a rectangular waveguide using a tunable cavity. By finding the frequencies at which the cavity resonates, we can identify the operating frequencies of the dominant $TE_{10}$ mode and verify the relationship between frequency, wavelength, and wave propagation velocities.

### Theory

Electromagnetic waves can only propagate through a rectangular waveguide if their frequency is above the cutoff frequency. For the dominant $TE_{10}$ mode, the cutoff frequency is $f_c = c/(2a)$ where $a$ is the larger dimension of the waveguide and $c$ is the speed of light. The phase velocity and group velocity for a wave traveling in the guide are given by:

$$v_p = \frac{c}{\sqrt{1-(f_c/f)^2}}, \qquad v_g = c\sqrt{1-(f_c/f)^2}$$

At resonance, the tunable cavity absorbs the most power, which shows up as a sharp dip on the SWR meter. Measuring this resonance frequency allows us to confirm the expected propagation behavior of the waveguide and compare it with theory.

### Setup

This setup is similar to Part 1, but now a tunable cavity (HP X532B frequency meter) is placed between the slotted waveguide carriage and the short-circuit termination.

**Steps:**

1. Remove the short-circuit termination from the carriage and attach it to the output of the tunable cavity.
2. Connect the frequency meter to the slotted waveguide while keeping the oscillator at 11 GHz.
3. Adjust the oscillator frequency and slowly tune the cavity until a sharp dip appears on the SWR meter, indicating resonance.
4. Read the resonant frequency directly from the frequency meter scale.
5. Repeat the measurements for 9.5 GHz and 8.5 GHz inputs.

### Equations Used

$f_c = c/(2a)$, where $c = 3 \times 10^8$ m/s (speed of light in vacuum) and $a = 22.92$ mm (width of the waveguide)

Phase Velocity: $v_p = \dfrac{c}{\sqrt{1-(f_c/f_{\text{measured}})^2}}$

Group Velocity: $v_g = c\sqrt{1-(f_c/f_{\text{measured}})^2}$ (equivalently $v_g = c^2/v_p$)

## Results

| Oscillator $f$ (GHz) | Measured $f$ (GHz) | $v_g$ (m/s) | $v_p$ (m/s) |
|---|---|---|---|
| 11 | 11.050 | $2.489 \times 10^8$ | $3.616 \times 10^8$ |
| 9.5 | 9.405 | $2.219 \times 10^8$ | $4.055 \times 10^8$ |
| 8.5 | 8.515 | $1.898 \times 10^8$ | $4.741 \times 10^8$ |

Table 2: Resonant cavity frequencies.

The measured resonance frequencies were slightly lower than the oscillator settings for two of three cases, which is expected due to small losses and calibration limits in the setup. As the operating frequency decreased, the group velocity also decreased, while the phase velocity increased, matching waveguide theory.

The results confirm the relationship $v_p v_g = c^2$, showing that energy travels more slowly near cutoff, while phase velocity becomes greater than the speed of light (without violating physics, since phase velocity does not represent energy flow).

Overall, the measurements matched the expected behavior of the $TE_{10}$ mode, with small variations likely caused by probe alignment or tuning sensitivity.

## 5 Part 1 Questions

### 1) Why are $\lambda$ and $\lambda_g$ different?

$\lambda_g$ is always longer than $\lambda$ because the wave inside the guide bounces obliquely between the walls instead of going straight down $z$. The free-space wavelength is just $\lambda = c/f$, and then the guide wavelength is $\lambda_g = \lambda/\sqrt{1-(f_c/f)^2}$ along the axis. For example at 11 GHz, $\lambda$ is about 27.3 mm and our measured $\lambda_g$ is 32.7 mm, which is roughly 20% longer. And as $f$ drops toward $f_c$, the bounce angle gets steeper, which stretches $\lambda_g$ out to infinity right at cutoff.

### 2) What electric field would the probe detect at the short-circuit end?

The probe would read zero because the boundary condition forces the tangential E-field to be zero at the conductor surface, which makes the short itself a node of the standing wave. Tangential E just means the part of the E-field that lies flat along the wall (parallel to the surface), and that's exactly what the probe measures. For example our data shows the closest detectable minimum sits a few half-wavelengths away from the short, which lines up with that node being the reference point.

### 3) Expected distance to the nearest detectable minimum (and comparison)

Minima repeat every $\lambda_g/2$ along the guide and the short itself is the first one, so the nearest detectable minimum is just $n \times (\lambda_g/2)$ for the smallest $n$ where the probe can reach.

| $f$ (GHz) | $\lambda_g/2$ (mm) | $n$ | Expected $d$ (mm) | Measured $d$ (mm) | % diff |
|---|---|---|---|---|---|
| 11 | 16.36 | 5 | 81.80 | 83.5 | 2.1% |
| 9.5 | 21.56 | 4 ($\approx$ 4.43) | 86.24 | 95.6 | 10.9% |
| 8.5 | 27.84 | 3 | 83.52 | 85.4 | 2.3% |

For example at 11 GHz the expected is 81.8 mm and we measured 83.5 mm, which is roughly 2% off. Same story at 8.5 GHz, about 2.3% off. The 9.5 GHz case is the outlier at 10.9%, which is probably because we read a different minimum than the one we counted, or the probe was a little off-center. So two of three lined up close, and the third one is consistent with a counting error.

### 4) Number of half-wavelengths between the nearest minimum and the short-circuit end

Just divide the measured distance by $\lambda_g/2$ and round to the nearest integer: $n = d_{\text{measured}}/(\lambda_g/2)$.

| $f$ (GHz) | Measured $d$ (mm) | $\lambda_g/2$ (mm) | $n = d/(\lambda_g/2)$ | Nearest int $n$ |
|---|---|---|---|---|
| 11 | 83.5 | 16.36 | 5.10 | 5 |
| 9.5 | 95.6 | 21.56 | 4.43 | 4 (with uncertainty) |
| 8.5 | 85.4 | 27.84 | 3.07 | 3 |

For example at 11 GHz, 83.5 mm / 16.36 mm comes out to 5.10, so that's 5 half-wavelengths. 8.5 GHz lands on 3.07 which is basically 3. The 9.5 GHz one is 4.43 which rounds to 4, but the fact that it is not close to an integer is what gave us the 10% error in Q3. So the pattern matches the $TE_{10}$ standing wave, where $E = 0$ at the short and the next zero shows up half a guide wavelength in.

## 5) Using $a$ and $b$ to compute cutoff modes

**(a)** For our measured $a = 22.92$ mm and $b = 10.29$ mm, the three lowest modes are $TE_{10}$ at 6.54 GHz, $TE_{20}$ at 13.09 GHz, and $TE_{01}$ at 14.58 GHz. The formula is:

$$f_c = f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

and the lowest non-zero $(m, n)$ combos win. In free space $1/\sqrt{\mu\varepsilon} = c = 3 \times 10^8$ m/s, so:

$$f_c = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \times 10^8}{2}\sqrt{\left(\frac{m}{22.92 \text{ mm}}\right)^2 + \left(\frac{n}{10.29 \text{ mm}}\right)^2}$$

| Mode | $(m, n)$ | $f_c$ formula | $f_c$ (GHz) |
|---|---|---|---|
| $TE_{10}$ | $(1, 0)$ | $c/(2a)$ | 6.54 (lowest) |
| $TE_{20}$ | $(2, 0)$ | $c/a$ | 13.09 |
| $TE_{01}$ | $(0, 1)$ | $c/(2b)$ | 14.58 |
| $TE_{11}/TM_{11}$ | $(1, 1)$ | $(c/2)\sqrt{1/a^2 + 1/b^2}$ | 15.98 (4th, not in top 3) |

TM modes need both $m$ and $n \geq 1$, so the lowest TM mode is $TM_{11}$ at 15.98 GHz, which is the 4th mode overall and not in the top three. And since $a/b = 22.92/10.29 = 2.23 \approx 2$, the ordering is the classic X-band setup $TE_{10} < TE_{20} < TE_{01}$.

**(b)** Only $TE_{10}$ propagates at all three test frequencies. The rule is $f > f_c$, and our operating range is 8.5 to 11 GHz which sits above $TE_{10}$ (6.54 GHz) but below $TE_{20}$ (13.09 GHz).

- 8.5–11 GHz $>$ 6.54 GHz $\Rightarrow$ $TE_{10}$ propagates
- 8.5–11 GHz $<$ 13.09 GHz $\Rightarrow$ $TE_{20}$ does NOT propagate
- 8.5–11 GHz $<$ 14.58 GHz $\Rightarrow$ $TE_{01}$ does NOT propagate

For example at 11 GHz we are at $f/f_c = 1.68$ for $TE_{10}$ which propagates fine, and the same 11 GHz is only 0.84 of the $TE_{20}$ cutoff so that one just dies off without propagating. So the guide stays single-mode the whole time.

**(c)** Solving $v_p = f\lambda_g = c/\sqrt{1 - (f_c/f)^2}$ for $f_c$:

$$f_c = f\sqrt{1 - \left(\frac{c}{f\lambda_g}\right)^2} = f\sqrt{1 - \left(\frac{\lambda}{\lambda_g}\right)^2}$$

So you back out the cutoff from each measurement and average.

| Frequency (GHz) | $\lambda$ (mm) | $\lambda_g$ (mm) | $f_c$ (GHz) |
|---|---|---|---|
| 11 | 27.27 | 32.73 | 6.08 |
| 9.5 | 31.58 | 43.12 | 6.47 |
| 8.5 | 35.29 | 55.68 | 6.57 |

We get 6.08, 6.47, and 6.57 GHz, averaging 6.37 GHz vs the theoretical 6.54 GHz, which is roughly 2.6% off. That is pretty close given the probe loading and the vernier scale being limited to about 0.1 mm precision.

# 6 Part 2 Questions

## Q1

*Using the measured values for frequency and wavelength, calculate the phase velocity of the wave: $v_p = \lambda_g f$. How close (i.e. percentage difference) is this to your values from Experiment 3.1? Based on this, how important is it to know the frequency accurately?*

| Frequency (GHz) | $v_p$ (m/s) (Part 1) | $v_p$ (m/s) (Part 2) | % Difference |
|---|---|---|---|
| 11 | $3.599 \times 10^8$ | $3.616 \times 10^8$ | 0.5% |
| 9.5 | $4.096 \times 10^8$ | $4.055 \times 10^8$ | 1.0% |
| 8.5 | $4.733 \times 10^8$ | $4.741 \times 10^8$ | 0.2% |

The phase velocities from Part 2 agree with Part 1 within 1.1% across all three frequencies. For example at 11 GHz, Part 1 gave $3.599 \times 10^8$ m/s and Part 2 gave $3.616 \times 10^8$ m/s, which is 0.5% off. The reason the agreement is so close is that both calculations use the same $\lambda_g$ average, which means the only real difference is whether you use the dial-set frequency or the cavity-measured one. And the cavity reads within 0.5% of the dial, so the carry-through is small.

For frequency accuracy, a 1% error in $f$ gives you a 1% error in $v_p$ directly, which is fine at this level. The bigger reason to know $f$ accurately is mode control. For example if the frequency drifted above the $TE_{20}$ cutoff at 13.09 GHz you would suddenly excite higher-order modes and the measurements would fall apart. So the cavity meter is mainly there to confirm we are still in single-mode $TE_{10}$ territory.

## Q2

*What happens to the group velocity $v_g$ as frequency gets closer to the cutoff frequency $f_c$? Can you offer some physical explanation for this?*

$v_g$ goes to zero as $f$ approaches $f_c$. From the equation $v_g = c\sqrt{1-(f_c/f)^2}$, the term inside the root collapses to zero when $f = f_c$. The measured data confirms this trend:

| $f$ (GHz) | $f_c$ (GHz) | $f/f_c$ | $v_g$ (m/s) |
|---|---|---|---|
| 11 | 6.54 | 1.68 | $2.501 \times 10^8$ |
| 9.5 | 6.54 | 1.45 | $2.197 \times 10^8$ |
| 8.5 | 6.54 | 1.30 | $1.902 \times 10^8$ |

For example at 8.5 GHz we measured $v_g = 1.902\times10^8$ m/s, and at 11 GHz it is up to $2.501\times10^8$ m/s, which lines up with $f/f_c$ going from 1.30 to 1.68. So the closer you push to cutoff, the slower energy moves down the guide.

About the physical picture, the $TE_{10}$ mode is like two plane waves bouncing off the broad walls at angle $\theta$ where $\sin\theta = f_c/f$. And as $f$ drops to $f_c$, $\theta$ goes to 90°, which means the waves bounce sideways with no net axial advance. So below cutoff the wave can't make forward progress at all and just decays exponentially down the guide.